Rational and Irrational Numbers
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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1. 8
2. $\frac{6}{25}$
3. 4.47
4. 15
5. 3.1, $\frac{16}{5}$, $\sqrt{11}$
6. $\frac{7}{11}$
7. 6.32
8. —
9. $\frac{5}{6}$
10. 6

Curriculum

Level: Grade 8 / Pre-Algebra
8.NS.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

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3. *If they got 10: halved the radicand instead of taking its square root, so sqrt(20) came out as 20/2*
7. *If they got 20: halved the radicand instead of taking its square root, so sqrt(40) came out as 40/2*

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1. Classify $\sqrt{64}$.

Step 1. 64 appears in the perfect-square row of the table ($8^2 = 64$), so the root comes out exactly.

Step 2. $\sqrt{64} = 8$, and $8 = \dfrac{8}{1}$ — a ratio of two whole numbers with a non-zero denominator. That is exactly the definition of rational. $\boxed{8}$

Classification: **rational**.

2. Terminating decimal to fraction.

Step 1. Read the decimal aloud: "twenty-four hundredths". The words give the fraction: $0.24 = \dfrac{24}{100}$.

Step 2. Reduce by the common factor 4: $\dfrac{24}{100} = \dfrac{6}{25}$. $\boxed{\frac{6}{25}}$

Classification: **rational**. Completed table row: 0.24 | $\frac{24}{100}$ | $\frac{6}{25}$ | yes. Every terminating decimal converts this way, so *every* terminating decimal is rational.

3. $\sqrt{20}$: locate, classify, approximate.

(a) From the table, $4^2 = 16$ and $5^2 = 25$. Since $16 < 20 < 25$, we get $4 < \sqrt{20} < 5$.
(b) 20 is not a perfect square, and no fraction squares to 20 either, so $\sqrt{20}$ is **irrational**: its decimal never ends and never repeats.
(c) $\sqrt{20} = 4.4721359\ldots$, which rounds to $\boxed{4.47}$
Why the rounded value is a different number: 4.47 terminates, so $4.47 = \frac{447}{100}$ is rational; $\sqrt{20}$ is not. The rounded value is a useful stand-in on a number line, never an equality.
Common wrong answer: 10 — halving the radicand instead of taking the square root.

4. Classify $\sqrt{225}$.

Step 1. Read the table: the last column gives $15^2 = 225$, so 225 is a perfect square.
Step 2. $\sqrt{225} = 15 = \dfrac{15}{1}$. $\boxed{15}$
Classification: **rational**. A root is rational precisely when the radicand is a perfect square (of a rational number) — being "a decimal that looks tidy" is not the test.

5. Order $\sqrt{11}$, 3.1, $\frac{16}{5}$.

Step 1. Put all three in decimal form. $\dfrac{16}{5} = 3.2$ exactly, and 3.1 is already a decimal.
Step 2. Bracket the root with the square table: $3^2 = 9$ and $4^2 = 16$, so $\sqrt{11}$ is between 3 and 4. Squaring the candidate settles it: $3.2^2 = 10.24 < 11$, so $\sqrt{11} > 3.2$.

$$\boxed{3.1 \; < \; \tfrac{16}{5} \; < \; \sqrt{11}}$$

Classification: $\sqrt{11}$ is the irrational one; 3.1 and $\frac{16}{5}$ are both rational. Note that the irrational number is not the largest "because it has a radical" — it lands where its size puts it.

6. Repeating decimal to fraction.

(a) A repeating decimal is rational because the repeat is a *pattern*, and any repeating pattern can be captured by a fraction. Only decimals that never end *and* never repeat escape being a ratio.
(b) Reduce $\dfrac{63}{99}$ by the common factor 9: $\dfrac{63}{99} = \dfrac{7}{11}$. $\boxed{\tfrac{7}{11}}$
Check: dividing 7 by 11 by hand returns the repeating block again. *Classification:* **rational**.

7. $\sqrt{40}$: locate, classify, approximate.

(a) Two entries from the square table bracket it: $6^2 = 36$ and $7^2 = 49$, and $36 < 40 < 49$, so $6 < \sqrt{40} < 7$.
(b) 40 is not a perfect square, so $\sqrt{40}$ is **irrational**.
(c) $\sqrt{40} = 6.3245553\ldots$, which rounds to $\boxed{6.32}$
Common wrong answer: 20 — halving 40 instead of taking its square root.

8. Is $\sqrt{3} = 1.73$?

Step 1. Test the claim by squaring the right-hand side: $1.73^2 = 2.9929$, which is less than 3. A number whose square is too small is itself too small.
Step 2. Therefore

$$\boxed{\sqrt{3} \; > \; 1.73}$$

What 1.73 *really is:* a rational two-decimal approximation of $\sqrt{3}$ — close enough for a sketch, but a different number. The true decimal $1.7320508\ldots$ never ends and never repeats.

9. Classify $\sqrt{\frac{25}{36}}$.

Step 1. The square root of a quotient is the quotient of the roots: $\sqrt{\dfrac{25}{36}} = \dfrac{\sqrt{25}}{\sqrt{36}}$.
Step 2. The table shows $5^2 = 25$ and $6^2 = 36$, so both roots are exact and the result is a ratio of whole numbers. $\boxed{\frac{5}{6}}$
Classification: **rational**. A radical sign does not make a number irrational — what matters is whether the root comes out exactly.

10. A product of two irrational numbers.

(a) Multiply under one radical: $\sqrt{12} \cdot \sqrt{3} = \sqrt{12 \cdot 3} = \sqrt{36}$, and 36 is a perfect square. $\boxed{6}$
(b) Expected sentence: the claim is **false**, and this is a counterexample — $\sqrt{12}$ and $\sqrt{3}$ are both irrational, yet their product is the rational number 6. One counterexample is enough to disprove an "always" statement. (Accept any answer that identifies the claim as false and points at this product as the reason.)